MVLU COLLEGE
AI PRACTICAL NO. 8

AIM:Adaboost Ensemble Learning

- Implement the Adaboost algorithm to create an ensemble of weak classifiers.
- Train the ensemble model on a given dataset and evaluate its performance.
- Compare the results with individual weak classifiers.

#CODE:

```
# PRACTICAL 8: ADABOOST ENSEMBLE LEARNING
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt

from google.colab import files

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import AdaBoostClassifier

from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report
)
# Upload and read dataset
files.upload()
df = pd.read_csv("College_Marks_Dataset.csv")
# Prepare data
X = df.drop(columns=["Student_ID", "Name", "Grade"])
y = df["Grade"]

pre = ColumnTransformer([
    ("cat", OneHotEncoder(handle_unknown="ignore"), ["Class"])
], remainder="passthrough")

# Split data
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.20, random_state=42, stratify=y
)

# INDIVIDUAL WEAK CLASSIFIER

weak = Pipeline([
    ("pre", pre),
    ("model", DecisionTreeClassifier(max_depth=1, random_state=42))
])

weak.fit(X_train, y_train)
weak_pred = weak.predict(X_test)
weak_acc = accuracy_score(y_test, weak_pred)

print("\n===== INDIVIDUAL WEAK CLASSIFIER =====")
print("Accuracy:", round(weak_acc * 100, 2), "%")
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, weak_pred))
print("\nClassification Report:")
print(classification_report(y_test, weak_pred))

# ADABOOST ENSEMBLE

ada = Pipeline([
    ("pre", pre),
    ("model", AdaBoostClassifier(
        estimator=DecisionTreeClassifier(max_depth=1),
        n_estimators=100,
        learning_rate=0.5,
        random_state=42
    ))
])

ada.fit(X_train, y_train)
ada_pred = ada.predict(X_test)
ada_acc = accuracy_score(y_test, ada_pred)

print("\n===== ADABOOST ENSEMBLE =====")
print("Training completed successfully.")
print("Accuracy:", round(ada_acc * 100, 2), "%")
```

```
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, ada_pred))
print("\nClassification Report:")
print(classification_report(y_test, ada_pred))

# ACTUAL VS PREDICTED

result = pd.DataFrame({
    "Actual Grade": y_test.values,
    "Weak Classifier": weak_pred,
    "AdaBoost": ada_pred
})

print("\n===== ACTUAL VS PREDICTED =====")
print(result.head(15))

# COMPARISON

comparison = pd.DataFrame({
    "Model": ["Weak Classifier", "AdaBoost"],
    "Accuracy": [weak_acc, ada_acc]
})

print("\n===== MODEL COMPARISON =====")
print(comparison)

print(
    "\nAdaBoost Improvement:",
    round((ada_acc - weak_acc) * 100, 2),
    "%"
)


# COMPARISON GRAPH

plt.figure(figsize=(7, 5))
plt.bar(comparison["Model"], comparison["Accuracy"])
plt.xlabel("Model")
plt.ylabel("Accuracy")
plt.title("Weak Classifier vs AdaBoost")
```

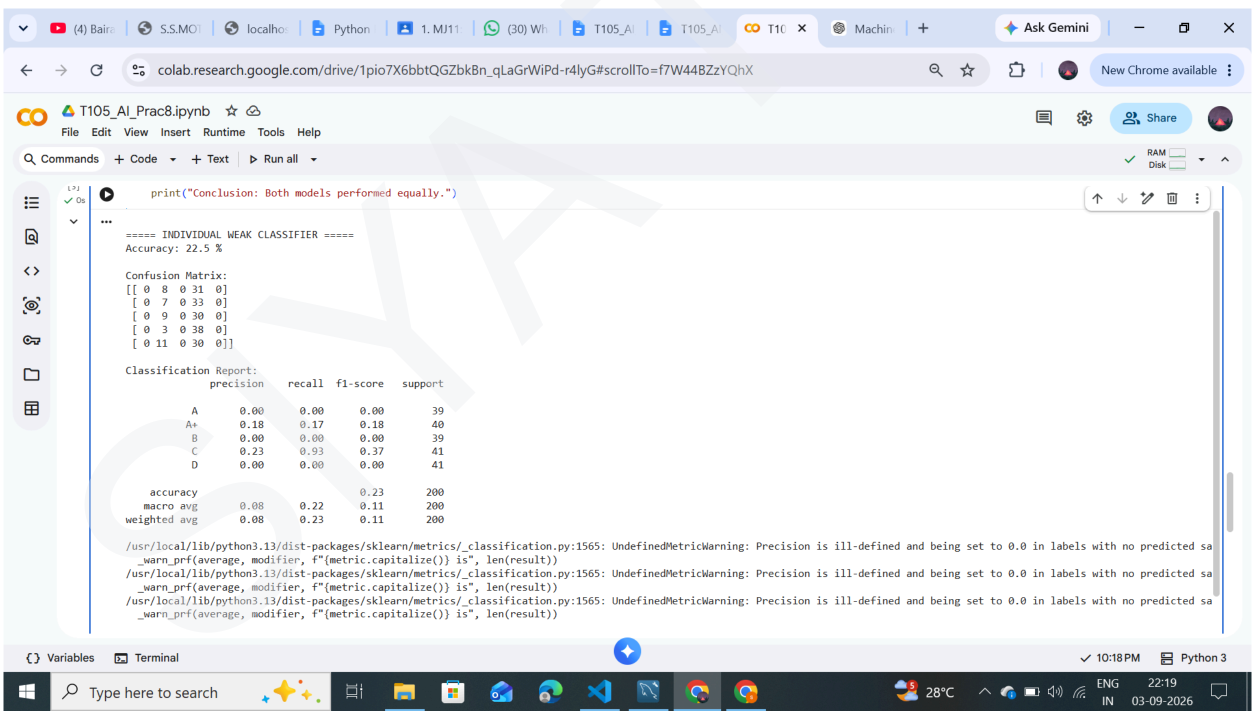
```python
plt.ylim(0, 1)
plt.show()

# FINAL RESULT

print("\n===== FINAL RESULT =====")
print("Weak Classifier Accuracy:", round(weak_acc * 100, 2), "%")
print("AdaBoost Accuracy:", round(ada_acc * 100, 2), "%")

if ada_acc > weak_acc:
    print("Conclusion: AdaBoost performed better.")
elif ada_acc < weak_acc:
    print("Conclusion: Weak Classifier performed better.")
else:
    print("Conclusion: Both models performed equally.")
```

#OUTPUT:



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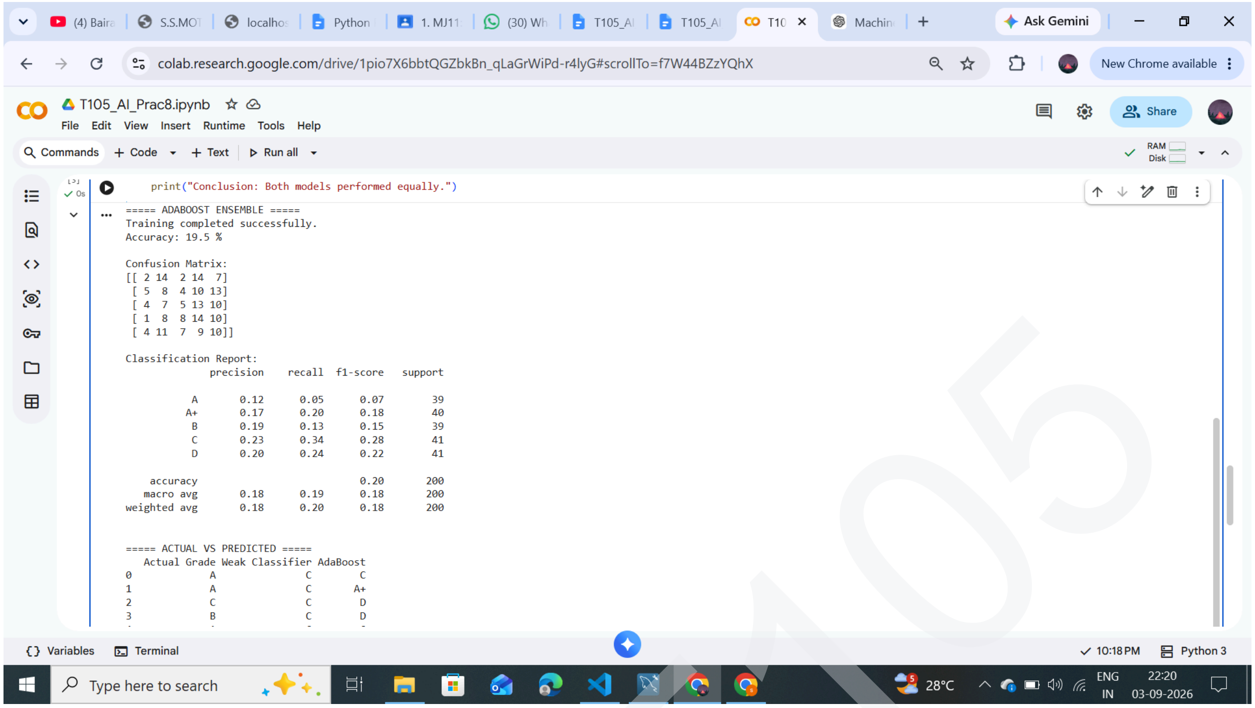
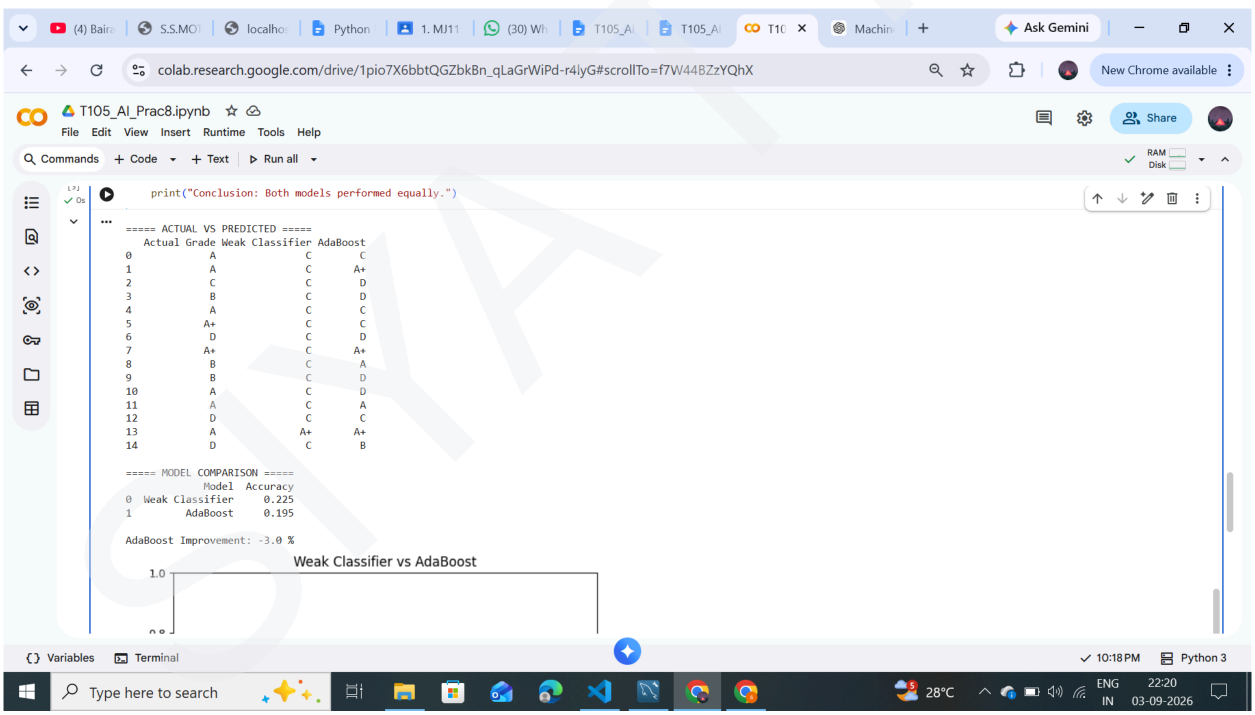


```
print("Conclusion: Both models performed equally.")

===== ADABOOST ENSEMBLE =====
Training completed successfully.
Accuracy: 19.5 %

Confusion Matrix:
[[ 2 14  2 14  7]
 [ 5  8  4 10 13]
 [ 4  7  5 13 10]
 [ 1  8  8 14 10]
 [ 4 11  7  9 10]]

Classification Report:
              precision    recall  f1-score   support

           A       0.12      0.05      0.07        39
          A+       0.17      0.20      0.18        40
           B       0.19      0.13      0.15        39
           C       0.23      0.34      0.28        41
           D       0.20      0.24      0.22        41

    accuracy                           0.20       200
   macro avg       0.18      0.19      0.18       200
weighted avg       0.18      0.20      0.18       200


===== ACTUAL VS PREDICTED =====
   Actual Grade Weak Classifier AdaBoost
0             A               C        C
1             A               C       A+
2             C               C        D
3             B               C        D
```



```
print("Conclusion: Both models performed equally.")

===== ACTUAL VS PREDICTED =====
   Actual Grade Weak Classifier AdaBoost
0             A               C        C
1             A               C       A+
2             C               C        D
3             B               C        D
4             A               C        C
5            A+               C        C
6             D               C        D
7            A+               C       A+
8             B               C        A
9             B               C        D
10            A               C        D
11            A               C        A
12            D               C        C
13            A              A+       A+
14            D               C        B

===== MODEL COMPARISON =====
             Model  Accuracy
0  Weak Classifier     0.225
1         AdaBoost     0.195

AdaBoost Improvement: -3.0 %
```

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